



REPORT

A lumped model with vadose filtering and quadratic drainage for daily groundwater recharge estimation

Caio Marques Manara¹ · Elias Hideo Teramoto^{1,2} · Gustavo Soldado Peres¹ · Pedro Luiz Becaro Crioni¹ · Hung Kiang Chang^{1,2,3}

Received: 19 November 2025 / Accepted: 30 June 2026
© The Author(s) 2026

Abstract

Quantifying groundwater recharge at a daily temporal resolution is difficult when inputs are sparse. This research presents a simple lumped model that links precipitation to groundwater levels by (1) routing rainfall through a first-order exponential vadose filter followed by an explicit discrete delay, and (2) representing lateral drainage with a nonlinear (quadratic) outflow law. The saturated storage is tracked with specific yield (S_y), and the model advances at a daily time step, producing time series of recharge, head, and lateral flow. The approach was tested with daily precipitation data from nearby rainfall stations and groundwater levels from four wells in the Bauru Aquifer System (BAS, Brazil). The model reproduces the timing and amplitude of observed hydraulic heads across seasonal cycles, capturing delayed and attenuated recharge pulses and curvilinear recessions, with a normalized root mean square error (RMSE) < 14% against observations. Relative to the water-table fluctuation (WTF) method, the mass balance explicitly accounts for concurrent lateral drainage during recharge events and for vadose delay/attenuation, explaining systematically higher, but process-consistent, recharge fractions than the WTF method. Parameter estimates are hydrologically interpretable, although equifinality requires validating fitted values against independent aquifer information. The method provides fast recharge diagnostics from long daily records without requiring daily evapotranspiration, making it suitable for basin-scale screening and trend attribution, and extensible to multistress forcing where data permit.

Keywords Groundwater recharge · Analytical modelling · Groundwater flow modelling · Bauru Aquifer System · Urucua Aquifer System

Introduction

Groundwater is the strategic reserve of the terrestrial water cycle, comprising 99% of accessible freshwater (UNESCO 2022). It sustains baseflow to rivers, buffers ecosystems and agriculture through droughts, and provides drinking-water supply for billions of people worldwide (e.g., Famiglietti

2014; Jasechko et al. 2024; UNESCO 2022; Soltanian et al. 2024). Effective management demands quantitative tools that map hydro-climatic forcing to aquifer response and to indicators across the socioeconomic, environmental and food and energy production systems (Ortiz-Partida et al. 2023). These tools must capture seasonal, spatial, and intensity-dependent variability and deliver fast, transparent, and defensible results (e.g., Samani 2021; Karimi et al. 2022; Taghilou 2022; Cuadrado-Quesada et al. 2024).

Groundwater recharge, the fraction of precipitation that percolates through the vadose zones and joins the saturated reservoir, is the pivotal process for replenishing groundwater (Healy 2010). Recharge controls the magnitude and timing of long-term water availability, water-table trends, and the resilience of exploitable water, thereby sustaining the agriculture and river baseflow during dry periods (e.g., Alam et al. 2022; Scanlon et al. 2023; Briggs et al. 2025). Consequently, quantifying recharge is critical for aquifer

✉ Caio Marques Manara
caio.m.manara@unesp.br

¹ Laboratory of Basin Studies (Lebac), São Paulo State University, Campus of Rio Claro, Rio Claro, São Paulo, Brazil

² Centre of Environmental Studies (CEA), São Paulo State University, Campus of Rio Claro, Rio Claro, São Paulo, Brazil

³ Department of Geology, São Paulo State University, Campus of Rio Claro, Rio Claro, São Paulo, Brazil

management; however, recharge is notoriously difficult to measure directly owing to its spatial heterogeneity and transient nature, and because it occurs below the land surface where observations are sparse (Healy and Cook 2002; Scanlon et al. 2002; Moeck et al. 2020).

A wide spectrum of approaches exists to estimate recharge, varying in scale, data demands, and underlying assumptions (e.g., Healy and Cook 2002; Scanlon et al. 2002; de Vries and Simmers 2002). Methods span surface-water and soil-water balances, tracer tests, hydrograph separation, and fully numerical models. Among these, techniques that exploit groundwater-level (GWL) fluctuations are particularly attractive where long daily well records exist: the aquifer acts as an integrator of climate forcing, and the input requirements are relatively modest. The classical water-table fluctuation (WTF) approach infers recharge from GWL rises given an assumed specific yield (S_y) (e.g., Águila et al. 2019; Labrecque et al. 2020; Gumuła-Kawęcka et al. 2022). Despite its simplicity, WTF is sensitive to the choice (and variability) of S_y , can misattribute overlapping events when storm sequences occur in quick succession, and often represents recessions poorly because it neglects delayed vadose transfer and contemporaneous lateral outflow (Healy and Cook 2002). Data issues, like barometric efficiency, tidal/earth-tide signals, pumping influences, and irregular sampling, further complicate event detection and bias specific yield (S_y)-based inferences if not preprocessed.

As an alternative, analytical lumped models can provide a useful intermediate framework between WTF methods and fully mechanistic simulators, often aiming to preserve a degree of hydrological interpretability in their parameters (e.g., Collenteur et al. 2021; Seidenfaden et al. 2023; Rudolph et al. 2023). These models retain simplicity but embed essential processes, yielding closed-form or very lightweight update equations that can be calibrated efficiently against multiyear time series. In doing so, they can partition observed head changes into recharge and lateral losses, reproduce curvilinear recessions without ad hoc corrections, and provide a full time series of recharge, heads, and drainage with quantified uncertainties, all while avoiding heavy parameterization. Crucially, parameters remain hydrologically interpretable, enabling consistency checks against independent aquifer knowledge. This balance of realism and parsimony makes lumped analytical models well suited for basin-scale screening, sensitivity analysis, and trend attribution where inputs are limited, yet robust diagnostics are needed.

Prior work has shown that such models can reproduce the rise of the water table during recharge episodes and its decline under lateral drainage, often deriving simplified responses from the Boussinesq framework or related linearizations (e.g., Park and Parker 2008; Cuthbert 2010; Liang and Zhang 2012; Collenteur et al. 2021; Guillaumot

et al. 2022). Analytical approaches to infer recharge from precipitation and GWL data span several families with distinct trade-offs.

Linear reservoir models (e.g., Park and Parker 2008) calibrate well on long daily records but conflate vadose attenuation and delay and assume linear lateral drainage. WTF-based recharge estimators (e.g., Cuthbert 2010) are operationally simple and transparent but hinge on uncertain, time-variable specific yield and struggle with delayed/attenuated inputs. Boussinesq-based formulations (e.g., Liang and Zhang 2012) allow nonlinear lateral outflow and capture peaks and recessions more realistically, yet require geometric assumptions and can face identifiability challenges.

Transfer-function/impulse-response (TFN) models (e.g., Collenteur et al. 2021) provide robust statistical fits via linear convolution of stresses but keep groundwater recharge implicit and handle discrete delays and strong nonlinear drainage less naturally. Recent extensions and hybrid schemes (e.g., Guillaumot et al. 2022) add vadose realism or multistress separation while retaining lumped storage, though many relinearize drainage or introduce additional parameters that complicate calibration.

Despite their efficiency, many existing formulations couple vadose-zone attenuation and delay into a single effective parameter and assume linear drainage, which can limit realism during strongly peaked events and high-head recessions. To address these gaps, this report proposes and tests a lightweight lumped recharge-water-level model extending the Park and Parker (2008) framework (1) explicitly separating vadose-zone smoothing (via a first-order exponential filter) from transport delay (via an explicit discrete lag), and (2) incorporating a nonlinear (quadratic) function to represent lateral drainage. Driven solely by daily precipitation, the resulting scheme links rainfall directly to GWL through a transparent daily mass balance. This yields a model that captures rapid rises and nonlinear recessions without requiring high-frequency evapotranspiration data, while remaining simple enough for basin-scale screening, automated calibration within physically bounded ranges, or manual trial-and-error tuning. The model returns diagnostic daily time series of recharge, lateral flow, and absolute head suitable for water-resources analysis.

Materials and methods

Framing

The model is conceptually based on the lumped-parameter, linear reservoir framework (e.g., Park and Parker 2008). In that foundational approach, the aquifer is treated as a single storage unit where recharge is often a direct fraction

of precipitation and lateral outflow is linearly proportional to the hydraulic head.

Two primary modifications were introduced that form the basis of the method: (1) a two-parameter filter is implemented to explicitly decouple precipitation smoothing (attenuation, κ) from transport time (delay, τ_v), and (2) a nonlinear (quadratic) function is used to represent the lateral drainage (outflow) process. This conceptual structure is illustrated in Fig. 1 and provides the basis for the mathematical formulation detailed in the following sections.

From precipitation to delayed recharge (vadose block)

Let $P_m(t)$ denote daily precipitation that infiltrates and travels through the unsaturated (vadose) zone. Attenuation during transport is represented by routing $P_m(t)$ through a first-order exponential filter, which is governed by the differential equation:

$$\frac{dr}{dt} = \frac{1}{\kappa} (P_m(t) - r(t)) \quad (1)$$

where κ (days) is the e-folding time, or smoothing constant and $r(t)$ the attenuated rain signal. For a daily time step ($\Delta t = 1\text{day}$), this continuous equation can be written as an exact discrete solution, where n represents the current time step (day) and $n-1$ represents the previous day. This solution is known as an exponential moving average (EMA) filter:

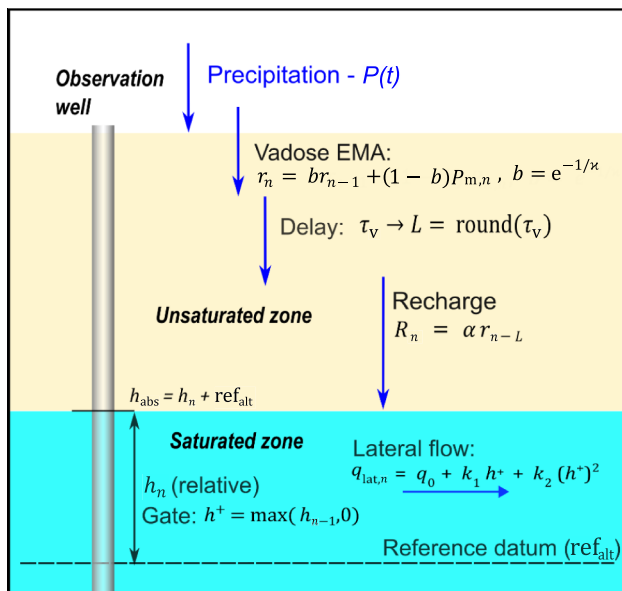


Fig. 1 Conceptual schematic of the lumped recharge–water-level model

$$r_n = br_{n-1} + (1-b)P_{m,n} \quad \text{where } b = e^{-\frac{1}{\kappa}} \quad (2)$$

A gain factor α (dimensionless) sets the fraction of the filtered signal that reaches the water table. Because percolation takes time, a discrete vadose delay τ_v (days) was imposed. Setting $L = \text{round}(\tau_v)$, the final delayed recharge R_n applied in the current time step to the saturated zone is:

$$R_n = \begin{cases} 0, & \text{if } n \leq L \\ \alpha r_{n-L}, & \text{if } n > L \end{cases} \quad (3)$$

To avoid start-up transients, the filter is seeded with an initial value r_0 derived from a chosen initial recharge value R_i :

$$r_0 = \frac{R_i}{\alpha} \quad (4)$$

Aquifer mass balance with explicit lateral drainage

The relative water level $h(\text{m})$ is defined with respect to a fixed datum. The lumped saturated storage per unit area is $S_y h$, where S_y (dimensionless) assumed constant. The aquifer water balance is thus:

$$S_y \frac{dh}{dt} = R(t) - q_{\text{lat}}(h) \quad (5)$$

The initial condition at the first post-delay time ($n = L$) is set as h_i . The model's net lateral outflow $q_{\text{lat}}(h)$ is defined explicitly with a quadratic law, using a gate function $h_n^+ = \max(h_{n-1}, 0)$ that prevents lateral inflow when the level is below initial condition:

$$q_{\text{lat},n} = q_0 + k_1 h^+ + k_2 (h^+)^2 \quad (6)$$

Here, $q_0(\text{m} \cdot \text{day}^{-1})$ is the background outflow, $k_1(\text{day}^{-1})$ sets the linear drainage strength, and $k_2(\text{m}^{-1} \cdot \text{day}^{-1})$ adds nonlinearity to capture stronger recessions at higher stages.

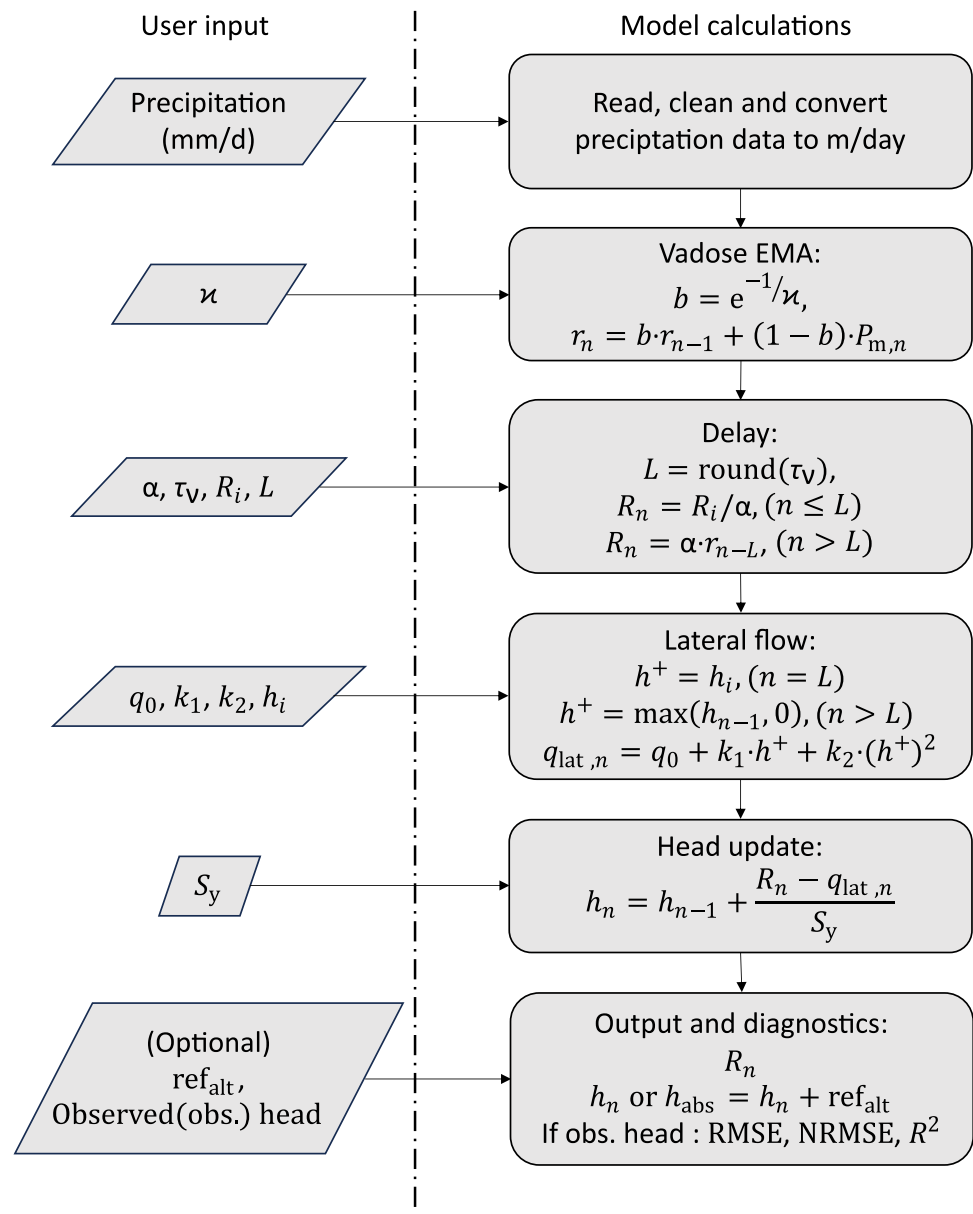
The model is advanced daily ($\Delta t = 1\text{day}$) using an explicit forward Euler scheme. The lateral flux for day n , $q_{\text{lat},n}$, is calculated based on the head from the previous day (h_{n-1}). The head h_n is then updated as:

$$h_n = h_{n-1} + \frac{R_n - q_{\text{lat},n}}{S_y} \Delta t \quad (7)$$

Algorithmic steps

The model implementation follows these traceable steps, following the workflow in Fig. 2:

Fig. 2 Workflow of the daily lumped recharge model. The left column lists user inputs; the right column shows the model calculations applied at each day n



1. *Input.* Read daily precipitation time series $P(t)$ and convert units to $P_m(t)$.
2. *Vadose EMA.* Compute $b = e^{-1/\alpha}$. Seed the filter with Eq. (4). Iterate Eq. (2) for all n to generate the r_n series.
3. *Discrete delay.* Set $L = \text{round}(\tau_v)$. Compute the final recharge series R_n using Eq. 3
4. *Head update.* Set initial head $h_L = h_i$. For all $n > L$, compute $q_{\text{lat},n}$ using Eq. (6) and advance h_n using Eq. (7).
5. *Diagnostics.* Export R_n , h_n , $q_{\text{lat},n}$. When observations exist, interpolate h_n to observation times and compute the root mean square error (RMSE) and coefficient of determination (R^2).

Study areas

Bauru Aquifer System

The Bauru Group covers a large region of southeastern Brazil, with an estimated area of 370,000 km², spanning parts of the states of Minas Gerais, Paraná, Mato Grosso do Sul, Goiás, and São Paulo. The geological framework of Bauru was shaped by a complex, rich history of predominantly siliciclastic sediment deposition in fluvial and eolian environments across the south-central sector of the Cretaceous South American Platform. This sedimentation occurred contemporaneously with alkaline magmatism at

several localities within the Paraná Sedimentary Basin, such as Lajes (SC), Iporá (GO), and Taiúva (SP). For the most part, sediments of the Bauru Group overlie the basalts of the Serra Geral Formation across an erosional contact (Brandt Neto et al. 1978; Paula e Silva et al. 2005; Menegazzo et al. 2016). In some regions, they rest upon units of the Guarani Aquifer System, the Botucatu and Piramboia formations. Its average thickness is approximately 100 m, reaching 300 m in the Marília area. Lithologically, it consists of sandstones, clayey sandstones, siltstones, mudstones, and claystones, with local conglomerates and limestone beds (Paula e Silva et al. 2005).

From a hydrodynamic standpoint, the Bauru Aquifer System (BAS) is a shallow, unconfined aquifer, with locally semiconfined portions. It exhibits high hydraulic conductivity, varying from 0.002 to 3.66 m/day, and relatively low storage capacity. Data compilations by CPRM (2012a) indicate that effective porosity typically ranges from 5 to 15%, but it can reach higher values (Bertolo 2001; Maldaner 2010). These characteristics make its water table highly responsive to short-term variations in recharge. Recharge occurs primarily through direct infiltration of rainfall, modulated by soil type, vegetation cover, and evapotranspiration dynamics. These features make the BAS an ideal setting to evaluate the capability of simplified analytical models to capture the temporal coupling between precipitation and GWL fluctuations.

Urucuia Aquifer System

The Urucuia Aquifer System (UAS) occupies an extensive area of central Brazil, predominantly in western Bahia and northern Minas Gerais, extending locally into Goiás and Tocantins. It is composed mainly of Upper Cretaceous siliciclastic sediments of the Urucuia Group deposited in fluvial–eolian environments on the São Francisco Craton. The aquifer is characterized by thick, laterally continuous quartz sandstones with subordinate siltstone and mudstone intercalations (Campos and Dardenne 1997; Gaspar et al. 2012).

The UAS directly overlies crystalline basement rocks and locally covers units of the Bambuí Group. Thickness varies significantly, commonly ranging from 100 to 300 m, but locally exceeding 500 m in depocenters (Gaspar et al. 2012; ANA 2017; CPRM 2018). The sediments are weakly consolidated and display high primary porosity and permeability owing to their quartz-rich composition and low diagenetic cementation.

Hydrogeologically, the UAS forms a regional unconfined aquifer with high transmissivity and storage capacity. Hydraulic conductivity values commonly range from 1 to 10 m/day, and effective porosity is typically between 10 and 25% (ANA 2017; CPRM 2018). Recharge occurs

mainly through direct infiltration of rainfall over extensive plateau surfaces, with groundwater flow directed toward deeply incised river valleys such as the São Francisco and its tributaries.

In contrast to the BAS, the UAS presents larger saturated thickness and longer groundwater residence times, resulting in smoother hydrographs and stronger buffering of short-term precipitation variability. However, intensive groundwater abstraction for large-scale irrigation has been associated with regional water-level declines and reduction of stored groundwater volume (e.g., Gonçalves et al. 2020a; Lucas et al. 2021; Camilo et al. 2024). These characteristics provide a complementary hydrogeological setting for testing the proposed formulation under conditions of higher storage and slower hydrodynamic response.

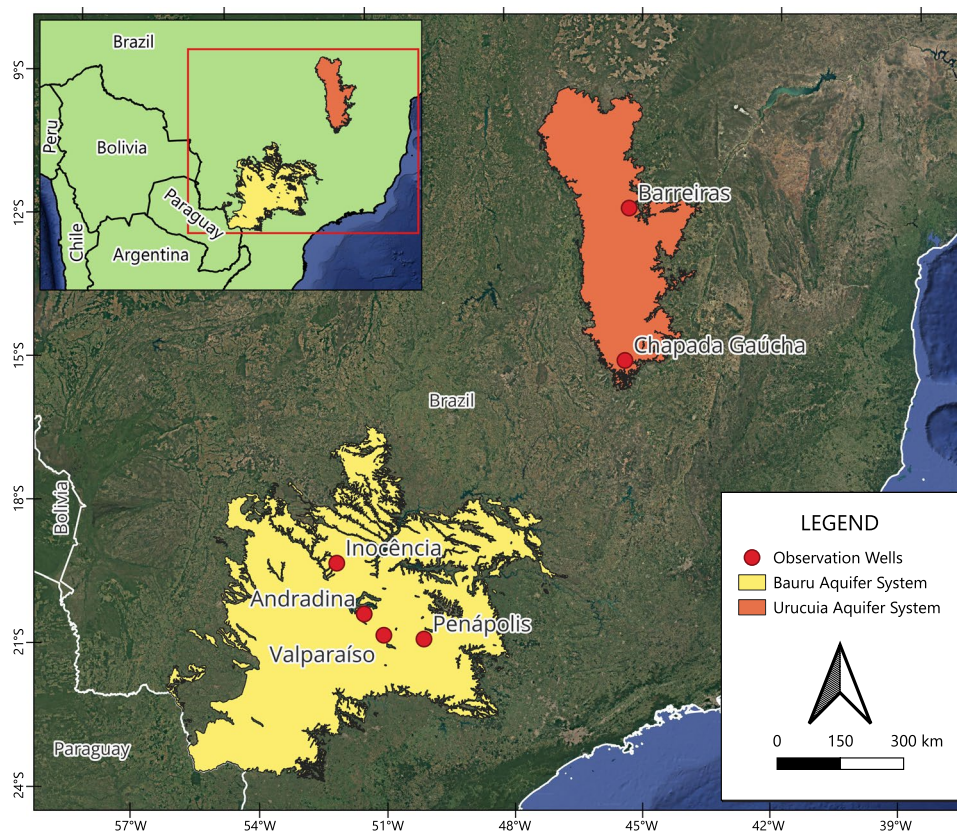
Selection of observation wells

To evaluate the applicability and performance of the proposed model in this setting, GWL data were obtained from the RIMAS database maintained by CPRM (Serviço Geológico do Brasil - Geological Survey of Brazil), which provides continuous monitoring records for wells across Brazil. Precipitation data were obtained from a pluviometric station operated by DAEE (Departamento de Águas e Energia Elétrica – São Paulo State Water and Energy Agency), with complementary hydrometeorological information provided by ANA (Agência Nacional de Águas e Saneamento Básico – National Water and Sanitation Agency). For this analysis, four monitoring wells from the BAS and UAS, located within 5 km of the precipitation station (Fig. 3), were selected, prioritizing those with the longest available time series and the fewest gaps in both precipitation and GWL records. This spatial proximity allows for direct comparability between the datasets and minimizes uncertainties associated with spatial variability in rainfall or aquifer response.

The simulation period corresponds to the available monitoring record of the CPRM network (starting in 2010). Because the dataset consists of continuous daily measurements, simulated and observed hydraulic heads were compared at each time step across the full time series, and performance metrics were calculated over the entire observation period.

The observation wells are installed in shallow, unconfined portions of both the BAS and UAS and are associated with localized recharge–discharge flow systems. Therefore, the monitoring network does not aim to provide spatially representative sampling of recharge at the scale of the entire aquifer. Instead, each well is treated as an independent local hydrodynamic response unit reflecting the recharge–discharge behavior of its respective small catchment. The large distances between wells correspond to different local

Fig. 3 Map of the study areas, illustrating the extent of the Bauru Aquifer System and Urucua Aquifer System. The locations of the six observation wells with nearby rain gauge stations (within 5 km) that provided the hydraulic head and precipitation data for the model simulations



catchments rather than to spatial continuity within a regional flow system.

Accordingly, the model is applied at the scale of the local hydrological catchment associated with each monitoring well, rather than at the scale of the BAS or the UAS as a whole. Although these aquifer systems are regionally extensive, the analyzed wells are screened within their shallow unconfined portions. Under these conditions, groundwater flow is predominantly local, and water-level fluctuations mainly respond to recharge–discharge processes occurring along short flow paths toward nearby drainage features. For this reason, precipitation measured at the nearest pluviometric station (located within 5 km of each well) is considered representative of the hydrological forcing controlling GWL dynamics at each site. Table 1 summarizes the geological characteristics and well completion details (total depth, diameter, water level, and screened interval) of the six selected monitoring wells in the BAS and UAS.

Model calibration and uncertainty assessment

Model calibration

Model calibration was performed by comparing simulated and observed hydraulic heads at a daily time step. The objective of the calibration procedure was to obtain a parameter

combination capable of reproducing both the magnitude and temporal dynamics of GWL fluctuations.

Calibration consisted of adjusting model parameters to minimize the RMSE between simulated and observed hydraulic heads while simultaneously maximizing the coefficient of determination (R^2).

Because GWL measurements are available at daily resolution, model performance was evaluated at each observation time step across the entire monitoring record. The calibration procedure therefore sought a parameter set capable of reproducing the complete hydrograph rather than isolated recharge events.

To avoid unrealistic solutions, the specific yield (S_y) was constrained within physically plausible ranges obtained from the literature. The resulting calibrated parameter set represents the best compromise between reproducing short-term fluctuations and long-term trends in groundwater levels.

Uncertainty analysis

To account for parameter uncertainty and the nonuniqueness of the calibration problem, a Bayesian Markov chain Monte Carlo (MCMC) framework was employed. This approach allows exploration of the parameter space and estimation of posterior probability distributions of model parameters conditional on the observed GWL data.

Table 1 Geological (lithology) and well completion details (total depth, diameter, depth to water range, and screened intervals) of the observation wells

ID	City	Aquifer system	General lithology	Final depth (m)	Diameter (cm)	Screened interval (m)	Depth to water range (m)
3500027324	Andradina, São Paulo State	BAS	Predominantly fine, subrounded quartz sand with minor silicification	60	15.24	36–56	10.76–16.91
3500027322	Penápolis, São Paulo State	BAS	Fine quartz sand with variable clay content	44	15.24	28–40	4.33–7.74
3500027323	Valparaíso, São Paulo State	BAS	Fine to very fine quartz sand with clayey matrix	52	15.24	32–48	7.21–11.52
3500034020	Inocência, Mato Grosso State	BAS	Fine- to medium-grained sandstone with variable carbonate cement	66	25.4	32–60	14.25–16.83
2900020686	Barreiras, Bahia State	UAS	Thick sequence of quartzose sandstones ranging from fine to coarse grained	72	10.16	0–46; 50–62; 66–70	11.99–13.29
3100020889	Chapada Gaúcha, Minas Gerais State	UAS	Interbedded fine- to medium-grained quartzose sand and sandy clay, with subordinate clayey sand layers	80	15.24	27.5–31.5; 39.5–47.5; 54.5–59.5; 67.5–71.5	6.93–9.41

According to Bayes' theorem, the posterior distribution of the parameter vector θ given observations y is presented in Eq. 8.

$$p(\theta | y) \propto p(\theta)p(y | \theta) \quad (8)$$

where $p(\theta)$ is the prior distribution and $p(y | \theta)$ is the likelihood function. Assuming independent and identically distributed Gaussian measurement errors, the likelihood function is defined in Eq. (9).

$$p(y | \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - f_i(\theta))^2}{2\sigma^2}\right) \quad (9)$$

where y_i are observed heads, $f_i(\theta)$ are simulated heads evaluated at observation times, and n is the number of observations. The residual standard deviation σ was estimated from the RMSE of the best-fit simulation.

Posterior sampling was performed using the affine-invariant ensemble sampler implemented in the *emcee* Python package (Foreman-Mackey et al. 2013), which employs multiple parallel walkers and is efficient for correlated parameter spaces. The sampler used 32 walkers and 5000 iterations, with the first 1000 iterations discarded as burn-in. Convergence was assessed by inspection of trace plots and by the Gelman–Rubin statistic ($\hat{R} < 1.1$ for all parameters).

Posterior samples were used to compute parameter summary statistics (mean, standard deviation, and 95% credible intervals), and parameter correlations were evaluated using corner plots. Prediction uncertainty was quantified by propagating parameter uncertainty through the model: 500 parameter sets were randomly drawn from the posterior distribution and used to rerun the model. The 2.5th and 97.5th percentiles of simulated heads define the 95% prediction interval. This Bayesian framework quantifies model reliability and distinguishes well-constrained parameters from those affected by equifinality inherent to the inverse problem.

Groundwater recharge assessment

Groundwater recharge was quantified exclusively from the daily recharge time series produced by the analytical lumped model. In the proposed formulation, recharge corresponds to the delayed and attenuated precipitation signal that reaches the water table after vadose-zone filtering and discrete transport lag. The model therefore provides a physically consistent daily recharge flux $R(t)$, expressed as an equivalent water layer. Annual recharge totals were computed by summing the simulated daily recharge over each hydrological year.

In most regions of Brazil, the climate exhibits marked dry–wet seasonality, with recharge occurring predominantly during the rainy season. Because recharge events may extend

across calendar-year boundaries, annual recharge and precipitation totals were computed using hydrological years defined from the onset of the rainy season. This procedure avoids splitting recharge pulses between two calendar years and ensures consistency with the seasonal recharge regime of the study area.

The use of hydrological years is particularly important in the present formulation, as vadose delay (τ_v) and smoothing (κ) distribute precipitation inputs over time. Consequently, rainfall occurring near the end of a calendar year may contribute to recharge during the subsequent months. Aligning the accounting period with the natural recharge cycle provides a more physically coherent representation of annual recharge efficiency and prevents artificial truncation of delayed recharge contributions.

Results

Simulation of water levels and groundwater recharge

The analytical model was capable of reproducing the observed hydraulic head fluctuations and its general trend across all four study sites. The model's performance was consistent, achieving a normalized root mean square error

(NRMSE), when compared to observed data, below 14% for all wells. The main products of the simulation for Inocência, Andradina, Penápolis, Valparaíso, Barreiras, and Chapada Gaúcha sites are presented in Figs. 4, 5, 6, 7, 8, and 9, respectively. To explicitly represent predictive uncertainty, 95% confidence bands were plotted around the simulated hydraulic head and recharge time series. These envelopes were derived from the Bayesian uncertainty analysis, in which parameter posterior distributions were propagated through the model. The shaded regions therefore reflect the 2.5th–97.5th percentile range of model realizations and provide a visual measure of the uncertainty associated with the estimated hydrographs and recharge fluxes.

The models' calibrated parameters showed a relatively narrow range of values: Specific yield (S_y) ranged from 0.14 to 0.16, the recharge coefficient (α) from 0.28 to 0.35 (28–35%), and the vadose delay (τ_v) from 27 to 35 days. The e-folding time (λ) showed the widest variability, ranging from 5 to 35 days (Table 2).

Furthermore, the parameters governing lateral flux (q_0 , k_1 , k_2) are highly interrelated, along with the initial head (h_i). It was noted during calibration that these parameters could be altered interchangeably while yielding very similar results, a common issue of equifinality. To manage this nonuniqueness, a simplified calibration approach was adopted, relying primarily on the constant flux parameter (q_0) to represent

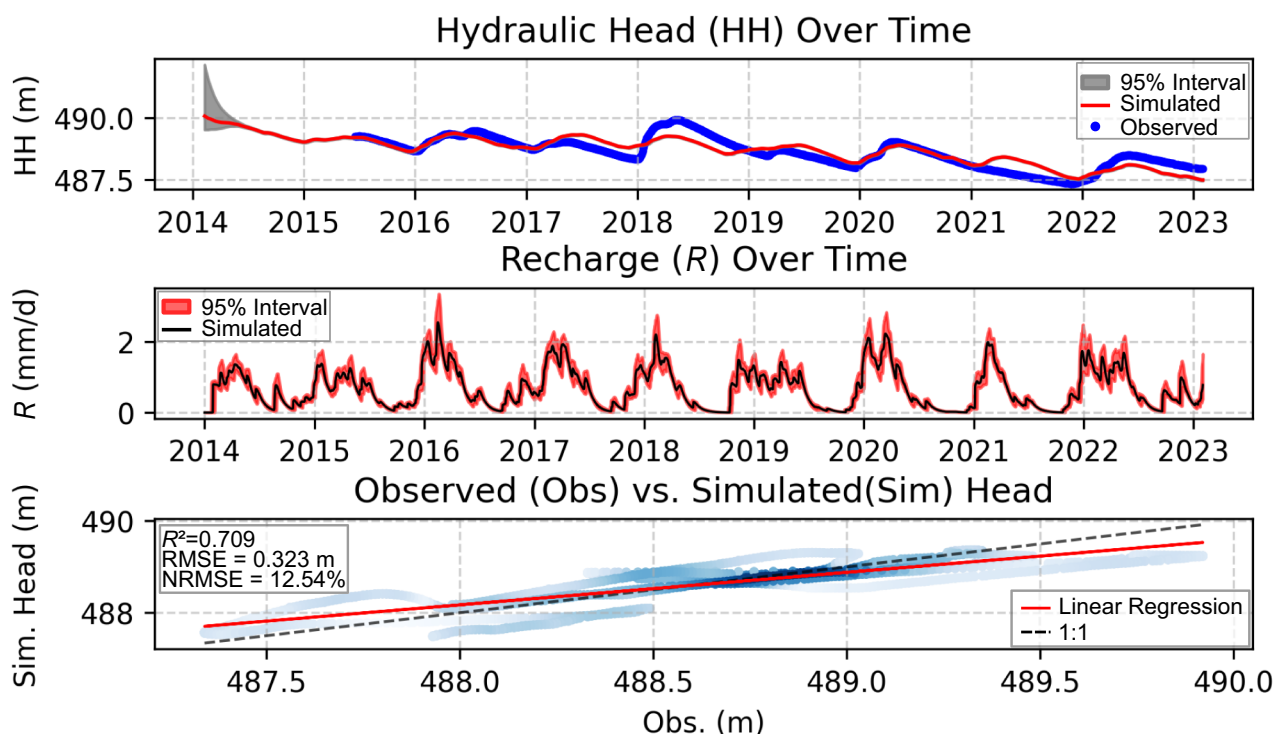


Fig. 4 Inocência site model results. The top panel presents the hydrograph comparison between simulated (Sim.) (red) and observed (Obs.) daily hydraulic head (blue), with the shaded envelope representing the 95% confidence interval derived from the Bayesian uncertainty analysis

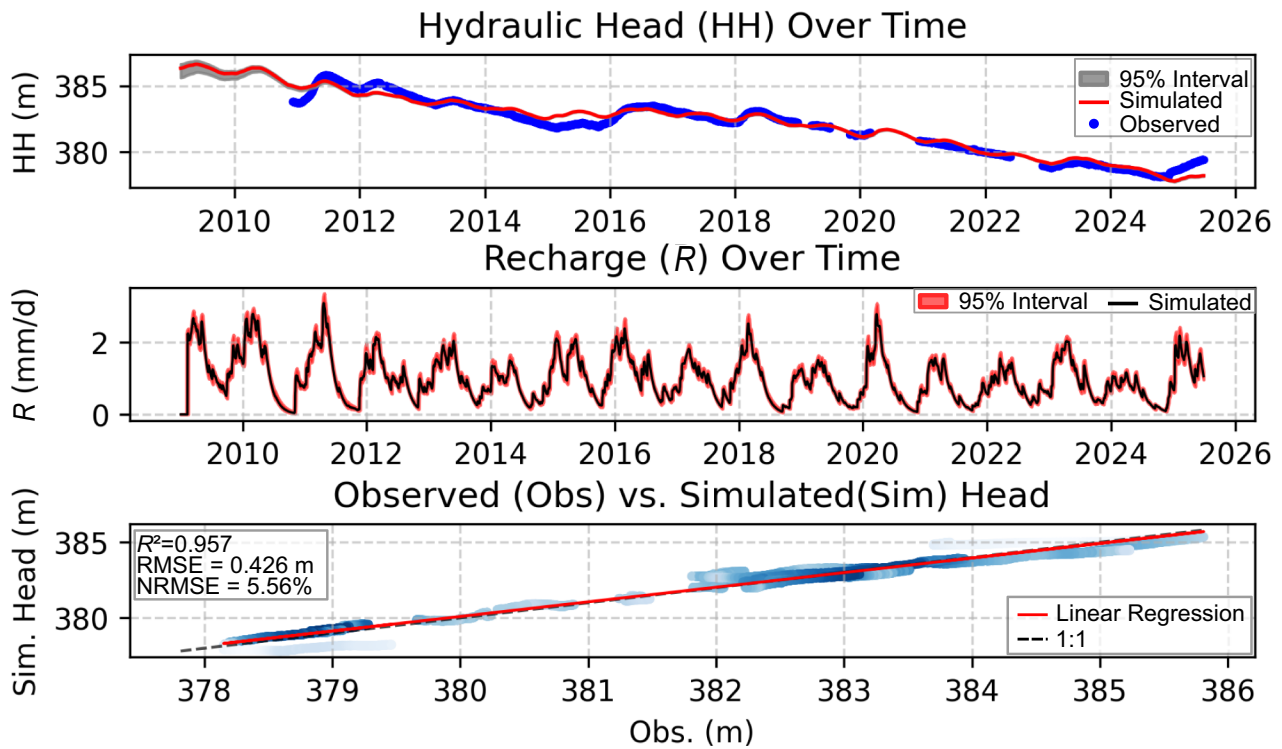


Fig. 5 Andradina site model results. The top panel presents the hydrograph comparison between simulated (Sim.) (red) and observed (Obs.) daily hydraulic head (blue), with the shaded envelope representing the 95% confidence interval derived from the Bayesian uncertainty analysis

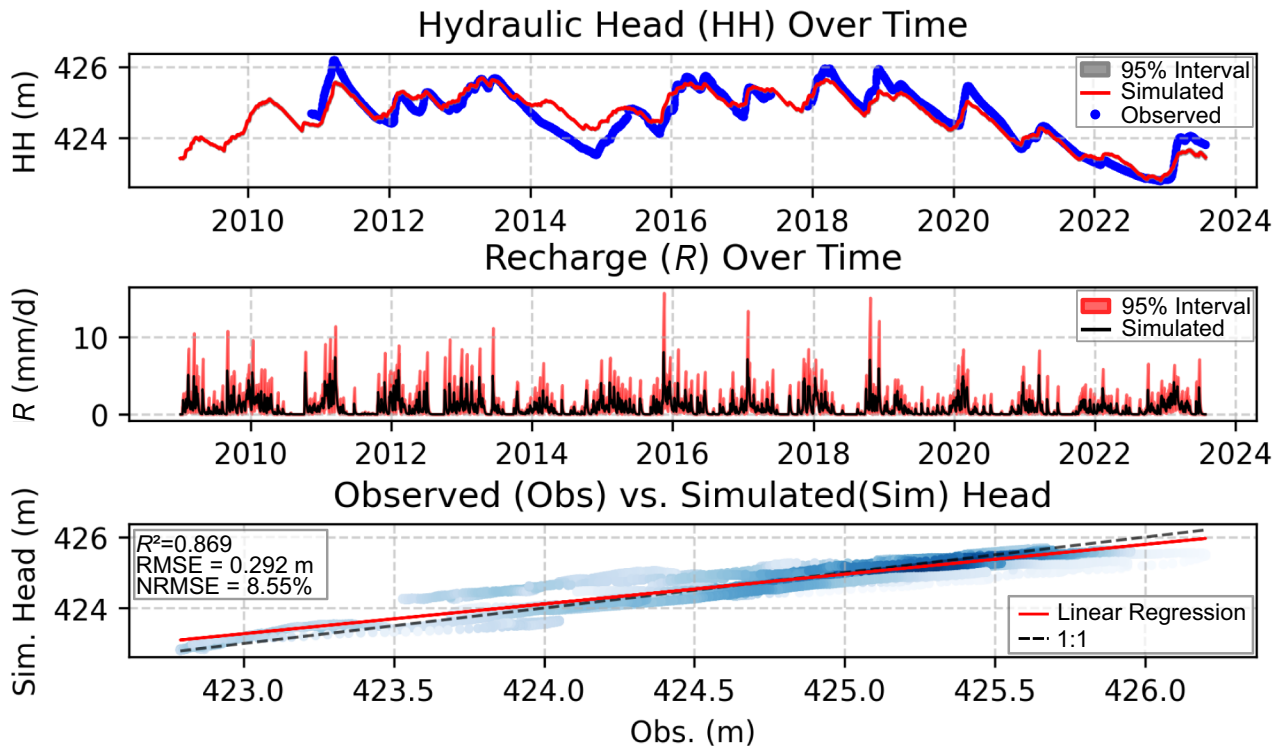


Fig. 6 Penápolis site model results. The top panel presents the hydrograph comparison between simulated (Sim.) (red) and observed (Obs.) daily hydraulic head (blue), with the shaded envelope representing the 95% confidence interval derived from the Bayesian uncertainty analysis

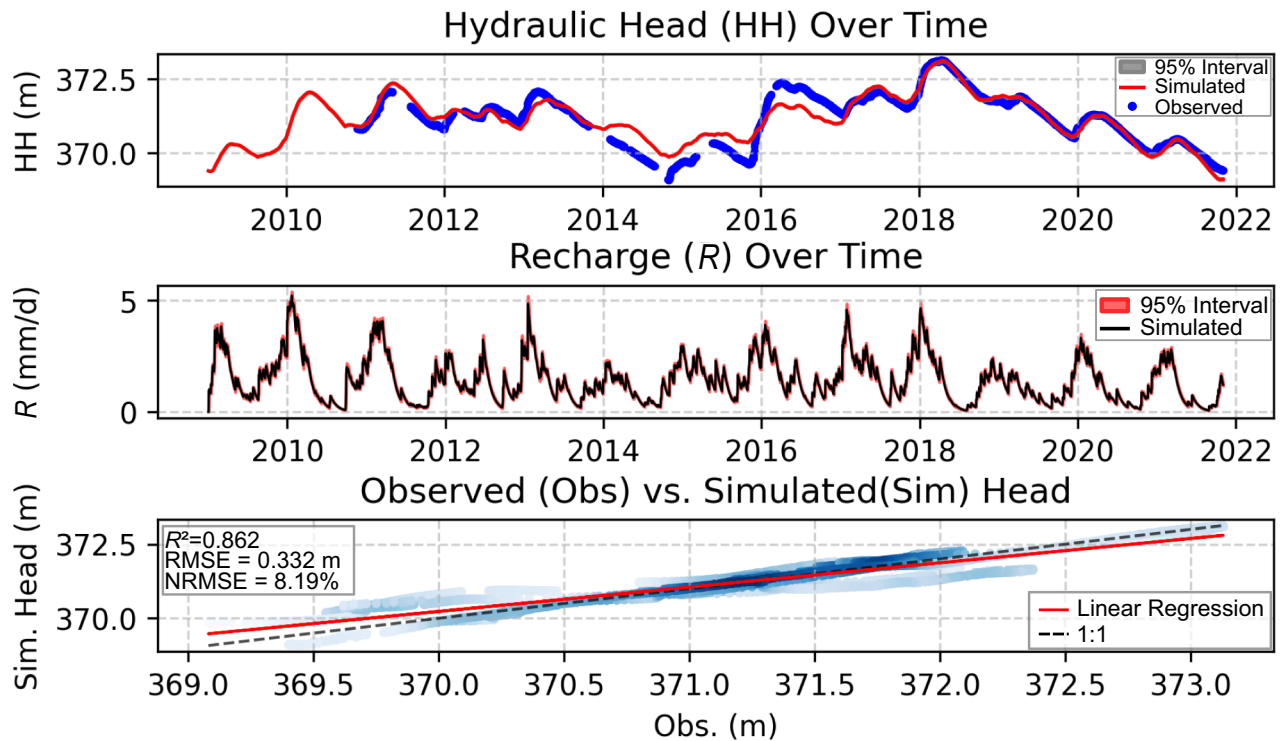


Fig. 7 Valparaíso site model results. The top panel presents the hydrograph comparison between simulated (Sim.) (red) and observed (Obs.) daily hydraulic head (blue), with the shaded envelope representing the 95% confidence interval derived from the Bayesian uncertainty analysis

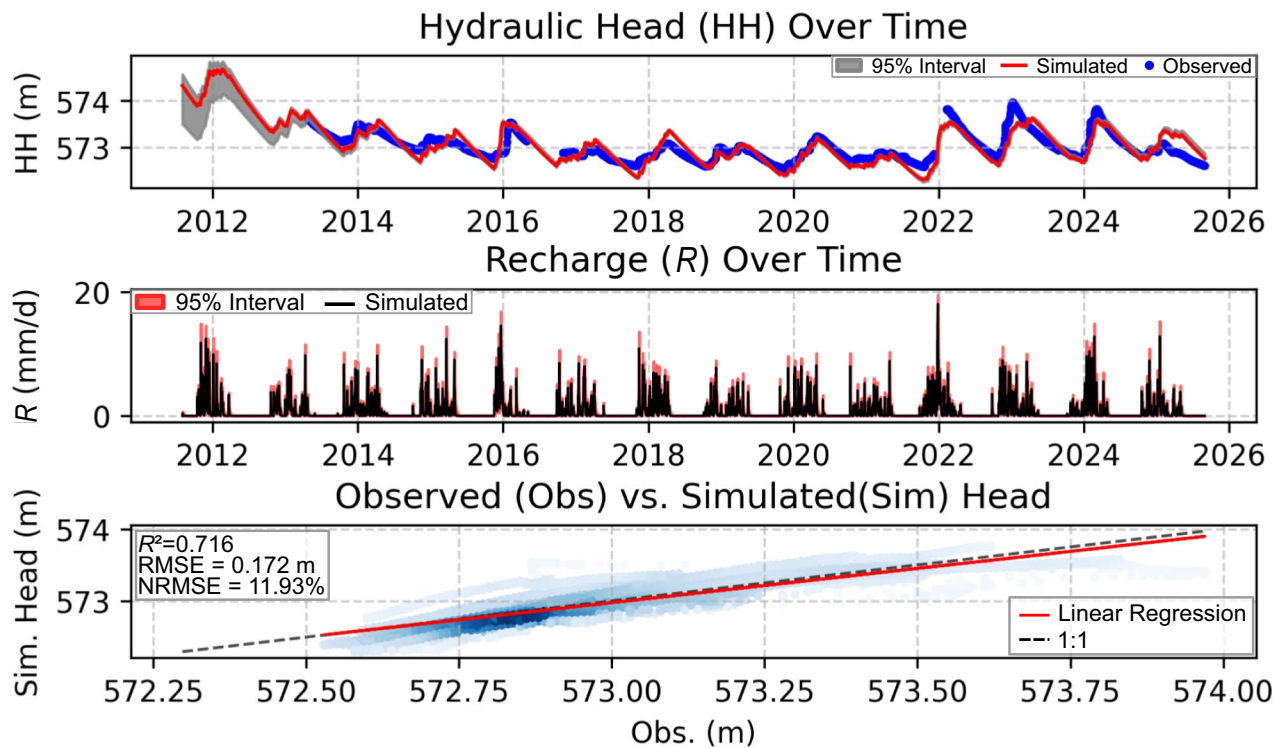


Fig. 8 Barreiras site model results. The top panel presents the hydrograph comparison between simulated (Sim.) (red) and observed (Obs.) daily hydraulic head (blue), with the shaded envelope representing the 95% confidence interval derived from the Bayesian uncertainty analysis

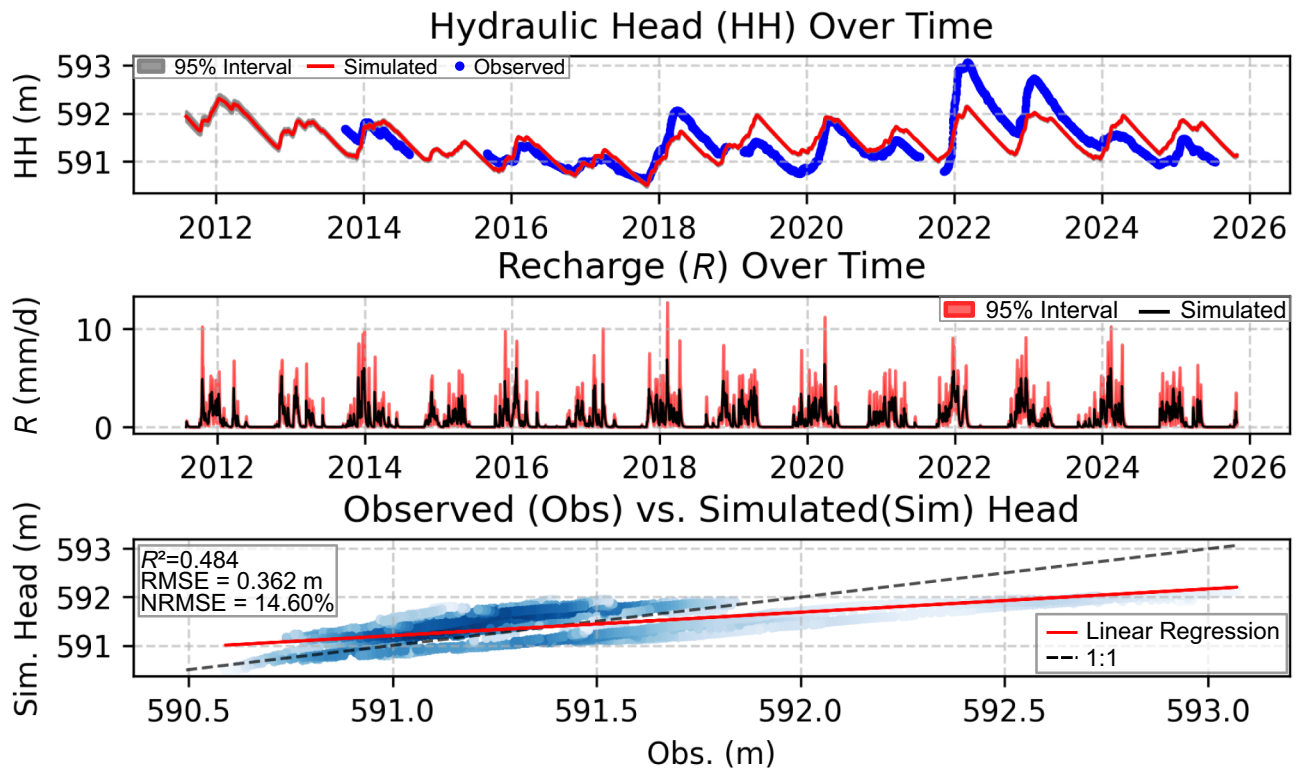


Fig. 9 Chapada Gaúcha site model results. The top panel presents the hydrograph comparison between simulated (Sim.) (red) and observed (Obs.) daily hydraulic head (blue), with the shaded envelope repre-

senting the 95% confidence interval derived from the Bayesian uncertainty analysis

Table 2 Calibration parameters of each observation wells

Parameters	Units	Inocência	Andradina	Valparaíso	Penápolis	Barreiras	Chapada Gaúcha
R_i	m/day	0.0008	0.002	0.001	0.001	0.0008	0.0008
κ	Days	22	41	31	36	1	5
α	Unitless	0.18	0.3	0.37	0.25	0.24	0.23
τ_v	Days	30	41	2	11	1	1
h_i	Meters	1.24	3.14	0	0	2.26	1.63
k_1	/day	2.52×10^{-3}	1.95×10^{-4}	4.68×10^{-5}	5.69×10^{-5}	2.59×10^{-4}	1.43×10^{-4}
k_2	m^{-1}/day	1.02×10^{-7}	7.74×10^{-8}	2.77×10^{-7}	1.75×10^{-7}	9.93×10^{-7}	4.84×10^{-7}
q_0	m/day	7.80×10^{-4}	1.11×10^{-3}	1.36×10^{-3}	7.83×10^{-4}	5×10^{-4}	5.46×10^{-4}
S_y	Unitless	0.15	0.16	0.14	0.15	0.16	0.16
ref_{alt}	Meters	489.5	383.2	369.4	423.4	572.2	590.3

the bulk of the net lateral outflow, and an initial condition (h_i) of zero. Despite this simplification, the calibrated models produced a simulated mean annual lateral flux ranging from 385 to 457 mm (expressed as an equivalent water layer) across the wells.

The resulting mean annual recharge estimates were 384.9 mm for Inocência, 320.4 mm for Andradina, 434.6 mm for Valparaíso, and 421.6 mm for Penápolis. It is worth noting that despite the analytical model using a constant α throughout the simulated time, effective annual

recharge-to-precipitation ratio (R/P) still exhibits a small variation when broken into smaller segments (annual in this case). This is an inherent result of the model's formulation, where the time lag (τ_v) and smoothing (κ) parameters distribute precipitation events over time. This process causes rainfall from the end of 1 year to sometimes contribute to the recharge of the next, thus preventing hard cut-offs between recharge cycles.

Estimated average groundwater recharge values derived from our simulations show clear spatial variability among

the analyzed wells (Table 3). Recharge ranges from 242.2 mm/year in Chapada Gaúcha and 247.0 mm/year in Barreiras to substantially higher values in Valparaíso (524.9 mm/year). Intermediate values are observed in Inocência (252.6 mm/year), Penápolis (289.1 mm/year), and Andradina (343.2 mm/year). Overall, the results indicate marked regional differences in recharge magnitude across the studied sites.

Discussion

The proposed model accurately reproduces both the long-term and cyclic variations, as well as the magnitude of observed hydraulic head fluctuations. It captures three hallmark features of groundwater hydrographs:

1. The phase lag between rainfall and water-table rise
2. Attenuation of event-scale peaks
3. The characteristically curved seasonal recession

These results indicate that the model is capable of representing recharge dynamics, providing daily estimates of recharge and lateral flow.

Several components of the present formulation, such as signal filtering, delayed response, and nonlinear recession behavior, are conceptually similar to structures used in transfer-function and other lumped hydrological models. The purpose of this study is not to introduce new mathematical operators, but to arrange these elements within an explicit groundwater mass-balance framework in which recharge is represented as a physical flux controlling GWL dynamics. The resulting formulation emphasizes physical interpretability and applicability under limited data availability rather than methodological novelty.

Hydrologically, the parameters are intuitive. The specific yield (S_y) controls the amplitude of water-level oscillations, while the vadose filter parameter (χ) controls the temporal spread of recharge pulses and thus event-scale attenuation. The delay term (τ_v) dictates the phase alignment of

the head response relative to precipitation, reflecting unsaturated-zone travel times. The coefficient α represents overall recharge efficiency, summarizing rapid losses such as surface runoff and near-surface evapotranspiration. Finally, k_1 , k_2 , and q_0 collectively define the baseflow level and the curvature of the recession limb.

The model does not explicitly simulate evapotranspiration (ET), which constitutes a deliberate simplification consistent with its intended application in data-limited settings where reliable ET time series are rarely available. Rather than prescribing ET as an independent flux, the formulation estimates an effective recharge fraction through the recharge–precipitation ratio (α). This parameter integrates the partitioning of precipitation at the land surface and within the vadose zone, such that only a portion of rainfall contributes to groundwater recharge while the remainder is dissipated through surface runoff and evapotranspiration. Consequently, ET losses are implicitly incorporated into α as part of the effective infiltration process. The model, therefore, estimates the effective recharge controlling GWL dynamics rather than attempting to resolve individual hydrological flux components. This parsimonious representation is consistent with the conceptual objective of capturing dominant hydrodynamic behavior under limited data availability while avoiding the introduction of poorly constrained parameters.

A key strength of this formulation is the explicit separation between vadose-zone smoothing (χ) and transport timing (τ_v), avoiding the common problem in transfer-function models where a single parameter must account for both effects. Moreover, the inclusion of a quadratic drainage term allows for nonlinear recession behavior, faster drainage at higher heads and slower at lower heads, providing additional flexibility even when not strictly required by the data.

Well-established hydrological models based on transfer function noise (TFN) formulations have demonstrated robust performance across diverse hydro-climatic settings (e.g., Fabbri et al. 2011; Collenteur et al. 2021; Rudolph et al. 2023). However, these approaches typically require evapotranspiration time series as a mandatory input, in addition to precipitation and discharge or GWL data. In many regions worldwide, including large parts of Brazil, long, continuous, and quality-controlled evapotranspiration observations remain scarce. Under such conditions, evapotranspiration must be estimated indirectly (e.g., through empirical relationships or energy-balance methods), introducing additional assumptions, parameters, and uncertainty sources that can obscure the interpretation of recharge processes.

The formulation proposed here was deliberately designed to operate under limited data availability, relying exclusively on precipitation and groundwater level records, which are more commonly monitored and accessible. This parsimony enhances applicability in data-scarce environments

Table 3 Estimated average groundwater recharge (mm/year) for the studied wells

Location	Average groundwater recharge (mm/year)
Inocência	252.6
Andradina	343.2
Penápolis	289.1
Valparaíso	524.9
Barreiras	247
Chapada Gaúcha	242.2

and allows a more direct interpretation of the relationship between recharge signals and water-table response.

Within the mass balance, precipitation pulses are delayed and attenuated while moving through the vadose zone. The recharge reaching the water table is then divided into an instantaneous lateral-flow component (q_{lat}) and a storage component that raises hydraulic heads. Consequently, GWL rise is controlled by the net input $R - q_{lat}$, whereas recession behavior reflects drainage dynamics.

Over long time scales, mean q_{lat} must approach mean recharge owing to mass conservation. The constant outflow term therefore controls the reference level, whereas the non-linear terms primarily influence the temporal structure of recession rather than the annual water balance.

To quantify these uncertainties, a Bayesian uncertainty analysis was performed, allowing exploration of parameter posterior distributions and propagation of parameter uncertainty into model predictions. In addition, to guard against compensating errors, parameter optimization during calibration was constrained using independent hydrogeological information and physically plausible bounds. In practice, S_y can be constrained by lithology and aquifer tests; κ and τ_v by soil-profile data, infiltration tests, or unsaturated-zone travel-time estimates based on depth to water; α by runoff coefficients and evapotranspiration budgets; and k_1 , k_2 , and q_0 by recession analysis of nearby streams, transmissivity ranges, and regional hydraulic gradients. These priors reduce the feasible parameter space and promote physically credible solutions.

The uncertainty envelopes derived from the posterior analysis indicate contrasting levels of confidence between simulated hydraulic heads and recharge estimates. In general, the variability associated with hydraulic-head simulations is very small, as evidenced by the narrow 95% confidence bands, suggesting that the model parameters are well constrained for reproducing water-level dynamics. In contrast, recharge uncertainty varies from well to well and can be substantial in specific cases. In the examples here, the estimated uncertainty was generally small but became significant in Inocência (Fig. 4), Penápolis (Fig. 6), and Chapada Gaúcha (Fig. 9), where the recharge confidence interval becomes comparatively wide during certain periods. Such differences reflect the inherent sensitivity and equifinality associated with recharge estimation in lumped approaches. These results indicate that, while hydraulic-head simulations are generally robust, recharge estimates should be interpreted with appropriate caution, particularly at sites where the uncertainty bands expand significantly. Whenever possible, information such as aquifer properties, water-level records, and auxiliary hydrological measurements should be used to constrain model inputs and reduce predictive uncertainty. Failure to adequately constrain inputs may lead to acceptable fits in hydraulic heads while masking substantial

uncertainty in inferred recharge fluxes. Calibrating the model to the full daily hydrograph, including both rises and recessions, enforces mass conservation and typically yields slightly higher but more defensible recharge fractions.

The estimated recharge values exhibit pronounced regional variability, reflecting differences in hydroclimatic conditions, aquifer properties, and local hydrological controls. Lower recharge rates in Chapada Gaúcha and Barreras contrast with the substantially higher value obtained for Valparaíso, while the remaining sites present intermediate behavior. Such spatial variability is expected in large sedimentary aquifer systems and reinforces the importance of site-specific parameterization when applying lumped recharge models. These differences also highlight that regional-scale generalizations of recharge may obscure important local controls on groundwater replenishment.

The approach outlined in this study, like other mathematical models, is subject to parameter nonuniqueness, whereby similar hydrograph fits can be obtained with different parameter combinations. Correlations among parameters exacerbate this issue. For example, S_y and α both influence amplitude; κ and τ_v both affect the temporal spreading and timing of recharge arrival; and k_1 , k_2 , and q_0 co-determine the recession shape and base level. When the forcing lacks sufficient variability (e.g., clustered storm sequences) or the observation record is short/noisy, these trade-offs yield broad, flat likelihood ridges and weak identifiability. So, to guard against compensating errors, the optimization of parameters during the model calibration values should be constrained by independent aquifer information and plausible bounds. In practice, S_y can be constrained by lithology and aquifer tests; κ and τ_v by soil-profile data, infiltration tests, or unsaturated-zone travel-time estimates based on depth to water; α by runoff coefficients and evapotranspiration budgets; k_1 , k_2 , and q_0 by recession analysis of nearby streams, transmissivity ranges, and regional gradients. These priors reduce the feasible parameter volume and promote physically credible solutions.

The model's performance declined during specific periods, reflecting processes not captured by the lumped formulation. In 2015, a pronounced regional drought (e.g., Figs. 6, 7 and 8) produced sharp water-level declines across all wells; steepened regional gradients and enhanced groundwater fluxes likely drove stronger declines than a fixed-parameter model could reproduce. Another discrepancy occurred at Inocência well during the 2017–2018 cycles: despite similar simulated recharge, the 2018 water-level rise was larger, even with slightly lower total precipitation (Table 2). This suggests a transient factor that increased the precipitation-to-recharge ratio and/or reduced effective porosity, possibly air entrapment temporarily lowering effective porosity (e.g., Faybishenko 1995; Gonçalves et al. 2020b; Ben-Noah et al. 2023; Iwagami et al. 2024), combined with differences in

intradaily rainfall timing or evapotranspiration that altered infiltration capacity. Recharge estimates obtained here align well with values reported for the BAS using various methods.

Recharge estimates agree with values reported for the BAS using multiple methods such as WTF, hydrologic budgets, and, more recently, GRACE-based approaches (e.g., Bertol et al. 2006; Bertol 2007; Maldaner 2010; Gonçalves et al. 2017; Ramires 2017; Zambianqui 2018; Gonçalves and Manzione 2019; Campos et al. 2021, 2023; Campos 2022). Reported recharge-to-precipitation ratios span 4–70% depending on method and location (Manzione et al. 2021), but cluster around a mean of ~26% and a median of ~24%, consistent with both analytical results and WTF estimates, with the former trending slightly higher and the latter trending slightly lower than the literature average.

The simulations are consistent with long-term GWL trends observed in shallow unconfined aquifers, which are strongly influenced by the temporal distribution of precipitation. In the BAS, which is intensively exploited in the State of São Paulo, the observed long-term decline likely reflects a combination of hydroclimatic and anthropogenic controls. In addition to the reduced rainfall and increased drought recurrence reported by Peres et al. (2025), groundwater abstraction from nearby pumping wells may contribute to the observed trends. In the current formulation, such persistent stresses are implicitly represented by the calibrated lateral drainage term (q_{lat}), allowing the model to reproduce the long-term declining behavior while strictly preserving the mass-balance structure. The good agreement obtained for the BAS suggests that the dominant drivers of the observed decline remain broadly compatible with the stationary effective outflow assumption.

For comparison, simulations were also performed for two wells located in the UAS, a region where large-scale irrigation and intensive groundwater abstraction have been widely associated with regional water-level declines and reductions in stored groundwater volume (e.g., Gonçalves et al. 2020a; Lucas et al. 2021; Renna Camacho et al. 2023; Camilo et al. 2024). Importantly, trend analyses for the UAS indicate no significant long-term decline in precipitation (Gonçalves et al. 2020a), suggesting that the reported storage depletion cannot be explained by rainfall forcing alone. In the UAS wells of this study (Figs. 8 and 9), the model captures the seasonal cyclicity and part of the long-term behavior through q_{lat} .

This growing mismatch is interpreted to reflect nonstationary anthropogenic stresses, most plausibly a progressive intensification of groundwater pumping, which cannot be fully represented when q_{lat} is treated as an effective stationary term. This limitation is not unique to the present formulation but is inherent to analytical lumped models in general, which are designed to represent aggregated system

behavior under quasi-stationary conditions. Under scenarios of increasing groundwater exploitation, such models may become progressively less able to reproduce long-term hydraulic-head declines. In this context, systematic residual trends not captured by the model can serve as a useful diagnostic indicator of emerging anthropogenic influence on the aquifer water balance.

Conclusions

The introduced and tested analytical framework maps daily precipitation to recharge and groundwater head by explicitly separating vadose-zone attenuation and delay from nonlinear lateral drainage. Across four wells in the Bauru Aquifer System, the model reproduced seasonal amplitudes, event timing, and recession curvature while yielding physically consistent daily series of recharge and lateral flow. Relative to the WTF method, the explicit mass balance naturally produces consistently higher, but mechanistically justified, recharge fractions by accounting for simultaneous drainage and vadose filtering. Because the parameters have clear hydrological meaning and calibration is fast with bounded search spaces, or can be tuned transparently by hand, the scheme is well-suited for basin-scale screening, trend attribution, and operational decision support in settings where daily evapotranspiration data are unavailable.

The main limitations arise from daily temporal resolution, implicit treatment of evapotranspiration, and parameter equifinality. These can be mitigated by imposing independent constraints on S_y , using longer observation records, and, where data permit, extending the forcing to include additional stresses. In practical terms, constraining S_y with lithologic and test data, bounding α and τ_v with soil-profile and travel-time information, and cross-checking recession parameters (k_1, k_2, q_0) against baseflow and transmissivity ranges improve identifiability and keep solutions physically credible. Calibrating to the full hydrograph (rises and recessions) helps enforce mass conservation, while cross-validation by hydrologic year tests transportability across wet and dry regimes.

The framework is readily extensible. Incorporating explicit evapotranspiration, pumping, or surface-water stage as additional forcings would allow attribution beyond precipitation-only control and reduce bias during droughts or managed-stress periods. Subdaily forcing could sharpen event timing in environments with flashy rainfall. Joint calibration across multiple wells, with gentle spatial regularization toward shared hydrofacies, can stabilize parameters and support map-based extrapolation of recharge indices. Where auxiliary observations are available, soil moisture, baseflow separation, or gravimetric water-storage tendencies can be

assimilated as additional calibration targets to further separate recharge from lateral losses.

Overall, the approach aims at a balance between realism and simplicity: It preserves process grounding through an explicit partitioning of vadose filtering, recharge, and drainage, yet remains parsimonious and computationally light. In data-constrained settings typical of regional aquifers, this balance enables recharge estimation and trend interpretation with minimal inputs, while leaving room for refinement when richer datasets become available.

Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were undertaken by Gustavo Soldado Peres, Pedro Luiz Becaro Crioni, Caio Marques Manara, and Elias Hideo Teramoto. The first draft of the manuscript was written by Caio Marques Manara and Elias Hideo Teramoto. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). The authors would like to thank the Fundação Para o Desenvolvimento da UNESP (FUNDUNESP) for granting research grants.

Data availability The data are available upon reasonable request.

Declarations Ethics approval and consent to participate:

Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this report.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Águila JF, Samper J, Pisani B (2019) Parametric and numerical analysis of the estimation of groundwater recharge from water-table fluctuations in heterogeneous unconfined aquifers. *Hydrogeol J* 27:1309–1328. <https://doi.org/10.1007/s10040-018-1908-x>
- Alam MF, Pavelic P, Villholth KG, Sikka A, Pande S (2022) Impact of high-density managed aquifer recharge implementation on groundwater storage, food production and resilience: a case from Gujarat, India. *J Hydrol Reg Stud* 44:101224. <https://doi.org/10.1016/j.ejrh.2022.101224>
- ANA – Agência Nacional de Águas e Saneamento Básico (2017) Estudos hidrogeológicos e de disponibilidade hídrica do Sistema Aquífero Urucuia e áreas adjacentes (Hydrogeological and water availability studies of the Urucuia Aquifer System and adjacent areas). ANA, Brasília
- Ben-Noah I, Friedman SP, Berkowitz B (2023) Dynamics of air flow in partially water-saturated porous media. *Rev Geophys* 61:e2022RG000798. <https://doi.org/10.1029/2022RG000798>
- Bertolo RA (2001) Hidrodinâmica e hidrogeoquímica da Zona não saturada do Aquífero Adamantina em Urânia – SP (Hydrodynamics and hydrogeochemistry of the unsaturated zone of the Adamantina Aquifer in Urânia – SP). PhD Thesis, University of São Paulo. <https://doi.org/10.11606/T.44.2001.tde-20012014-174523>
- Bertol GA (2007) Avaliação da recarga aquífera anual no aquífero Bauru no Município de Araguari, Minas Gerais (Evaluation of the annual groundwater recharge in the Bauru aquifer in the Municipality of Araguari, Minas Gerais). MSc Thesis, Universidade Federal de Minas Gerais. <http://hdl.handle.net/1843/MPBB-74MN3Y>. Accessed 7 July 2026
- Bertol GA, Camargos CC, Velásquez LNM, de Aquino Branco OE, Cota SD S (2006) Aplicação do método do cálculo do balanço hídrico na avaliação da recarga no Aquífero Bauru em Araguari – MG (Application of the water balance calculation method in evaluating the recharge of the Bauru Aquifer in Araguari – MG). Anais do XIV Congresso Brasileiro de Águas Subterrâneas, Brasília, 2006
- Brandt Neto M, Petri S, Coimbra AM (1978) Considerações sobre a estratigrafia e ambiente de sedimentação da Formação Bauru (Considerations on the stratigraphy and depositional environment of the Bauru Formation). Anais do XXX Congr Bras de Geol 2:557–566
- Briggs MA, Newman CP, Benton JR, Rey DM, Konrad CP, Ouellet V, Torgersen CE, Gruhn LR, Fleming BJ, Gazoorian CL, Doctor DH (2025) James Buttle review: the characteristics of baseflow resilience across diverse ecohydrological terrains. *Hydrol Process* 39(3):e70101. <https://doi.org/10.1002/hyp.70101>
- Camilo BK, Andrade PL, Eleutério JC, Rodrigues AF (2024) The importance of local observation wells, reanalysis, and satellite data on gravity anomaly, climate, and land use to improve groundwater management in the Urucuia aquifer system. *J South Am Earth Sci* 143:105018. <https://doi.org/10.1016/j.jsames.2024.105018>
- Campos JCV (2022) Interação das águas subterrâneas/superficiais: um paradigma na gestão dos recursos hídricos–Sistema Aquífero Bauru no Triângulo Mineiro-MG (Groundwater/surface water interaction: a paradigm in water resources management–Bauru Aquifer System in the Triângulo Mineiro-MG). PhD Thesis, Universidade Estadual Paulista. <http://hdl.handle.net/11449/236190>. Accessed 7 Jul 2026
- Campos JCV, Gastmans D, Santarosa LV, Lopes IO, Alves MCP, Martins LF, Fabian AJ (2021) Estimativas da Recarga do Sistema Aquífero Bauru em uma Microbacia no Entorno da Cidade de Uberaba – MG (Recharge estimates of the Bauru Aquifer System in a watershed surrounding the city of Uberaba – MG). *Anu Inst Geoci* 44:40585. https://doi.org/10.11137/1982-3908_2021_44_40585
- Campos JCV, Gastmans D, Santarosa LV (2023) Utilização do GRACE no cálculo da recarga das águas subterrâneas no Sistema Aquífero Bauru na Bacia do Rio São Jerônimo - Triângulo Mineiro (Use of GRACE to calculate groundwater recharge in the Bauru Aquifer System in the São Jerônimo River Basin - Triângulo Mineiro).

- Águas Subterr 37(2):e30168. <https://doi.org/10.14295/ras.v37i2.30168>
- Campos JEG, Dardenne MA (1997) Origem e evolução tectono-sedimentar da Bacia Sanfranciscana (Origin and tectono-sedimentary evolution of the Sanfranciscana Basin). *Rev Bras Geoci* 27(3):283–294
- Collenteur RA, Bakker M, Klammler G, Birk S (2021) Estimation of groundwater recharge from groundwater levels using nonlinear transfer function noise models and comparison to lysimeter data. *Hydrol Earth Syst Sci* 25(5):2931–2949. <https://doi.org/10.5194/hess-25-2931-2021>
- CPRM – Serviço Geológico do Brasil (2012) Projeto Rede Integrada de Monitoramento das Águas Subterrâneas: relatório diagnóstico sistema aquífero Guarani nos estados de São Paulo, Mato Grosso do Sul e Paraná bacia sedimentar do Paraná (Integrated Groundwater Monitoring Network Project: diagnostic report of the Guarani aquifer system in the states of São Paulo, Mato Grosso do Sul, and Paraná, Paraná sedimentary basin). CPRM, São Paulo
- CPRM – Serviço Geológico do Brasil, (2018) Sistema Aquífero Urucua: caracterização hidrogeológica e potencial hídrico (Urucua Aquifer System: hydrogeological characterization and water potential). CPRM, Belo Horizonte, Brazil
- Cuadrado-Quesada G, Lictevout E, Schmeier S, Vargas CR (2024) Revisiting groundwater law through the lenses of earth system law and rights of nature. *Wiley Interdiscip Rev Water* 11(2):e1684. <https://doi.org/10.1002/wat2.1684>
- Cuthbert MO (2010) An improved time series approach for estimating groundwater recharge from groundwater level fluctuations. *Water Resour Res* 46(9):W09515. <https://doi.org/10.1029/2009WV0008572>
- de Vries JJ, Simmers I (2002) Groundwater recharge: an overview of processes and challenges. *Hydrogeol J* 10(1):5–17. <https://doi.org/10.1007/s10040-001-0171-7>
- Fabbri P, Gaetan C, Zangheri P (2011) Transfer function-noise modeling of an aquifer system in NE Italy. *Hydrol Process* 25(2):194–206. <https://doi.org/10.1002/hyp.7831>
- Famiglietti JS (2014) The global groundwater crisis. *Nat Clim Change* 4:945–948. <https://doi.org/10.1038/nclimate2425>
- Faybishenko BA (1995) Hydraulic behavior of quasi-saturated soils in the presence of entrapped air: laboratory experiments. *Water Resour Res* 31(10):2421–2435. <https://doi.org/10.1029/95WR01654>
- Foreman-Mackey D, Hogg DW, Lang D, Goodman J (2013) emcee: the MCMC hammer. *Publications of the Astronomical Society of the Pacific* 125(925):306–312. <https://doi.org/10.1086/670067>
- Gaspar MTP, Campos JEG, Dardenne MA (2012) Estratigrafia, sedimentologia e evolução hidrogeológica do Grupo Urucua, oeste da Bahia (Stratigraphy, sedimentology, and hydrogeological evolution of the Urucua Group, western Bahia). *Rev Bras Geoci* 42(1):168–182. <https://doi.org/10.5327/Z0375-7536201200010012>
- Gonçalves RD, Stollberg R, Weiss H, Chang HK (2020a) Using GRACE to quantify the depletion of terrestrial water storage in northeastern Brazil: the Urucua Aquifer System. *Sci Total Environ* 705:135845. <https://doi.org/10.1016/j.scitotenv.2019.135845>
- Gonçalves RD, Teramoto EH, Engelbrecht BZ, Alfaro Soto MA, Chang HK, van Genuchten MT (2020b) Quasi-saturated layer: implications for estimating recharge and groundwater modeling. *Groundwater* 58(3):432–440. <https://doi.org/10.1111/gwat.12916>
- Gonçalves VFM, Manzione RL, Santarosa LV, Nava A (2017) Estimativa da recarga para o sistema Aquífero Bauru no Município de Assis/SP (Recharge estimation for the Bauru Aquifer System in the Municipality of Assis/SP). *Anais do XIX Congresso Brasileiro de Águas Subterrâneas*. <https://doi.org/10.14295/ras.v0i0.28726>
- Gonçalves VFM, Manzione RL (2019) Estimativa da recarga das águas subterrâneas no sistema Aquífero Bauru (SAB) (Groundwater recharge estimation in the Bauru Aquifer System [BAS]). *Geo UERJ* 35:e37063. <https://doi.org/10.12957/geouerj.2019.37063>
- Guillaumot L, Longuevergne L, Marçais J, Lavenant N, Bour O (2022) Frequency domain water table fluctuations reveal impacts of intense rainfall and vadose zone thickness on groundwater recharge. *Hydrol Earth Syst Sci* 26(22):5697–5720. <https://doi.org/10.5194/hess-26-5697-2022>
- Gumula-Kawęcka A, Jaworska-Szulc B, Szymkiewicz A, Górczewska-Langner W, Pruszkowska-Caceres M, Angulo-Jaramillo R, Šimůnek J (2022) Estimation of groundwater recharge in a shallow sandy aquifer using unsaturated zone modeling and water table fluctuation method. *J Hydrol* 605:127283. <https://doi.org/10.1016/j.jhydrol.2021.127283>
- Healy RW (2010) Estimating groundwater recharge. Cambridge University Press, Cambridge
- Healy RW, Cook PG (2002) Using groundwater levels to estimate recharge. *Hydrogeol J* 10:91–109. <https://doi.org/10.1007/s10040-001-0178-0>
- Iwagami S, Noguchi S, Shimizu T, Kubota T, Iida S (2024) Field measurement of entrapped pore-air pressure and the effect of rising groundwater level in the soil layer. *Hydrol Process* 38(7):e15235. <https://doi.org/10.1002/hyp.15235>
- Jasechko S, Seybold H, Perrone D, Fan Y, Shamsudduha M, Taylor RG, Fallatah O, Kirchner JW (2024) Rapid groundwater decline and some cases of recovery in aquifers globally. *Nature* 625(7996):715–721. <https://doi.org/10.1038/s41586-023-06879-8>
- Karimi D, Bahrami J, Mobaraki J, Missimer TM, Taheri K (2022) Groundwater sustainability assessment based on socio-economic and environmental variables: a simple dynamic indicator-based approach. *Hydrogeol J* 30:1963–1988. <https://doi.org/10.1007/s10040-022-02512-6>
- Labrecque G, Chesnaux R, Boucher MA (2020) Water-table fluctuation method for assessing aquifer recharge: application to Canadian aquifers and comparison with other methods. *Hydrogeol J* 28:521–533. <https://doi.org/10.1007/s10040-019-02073-1>
- Liang X, Zhang Y (2012) A new analytical method for groundwater recharge and discharge estimation. *J Hydrol* 450:17–24. <https://doi.org/10.1016/j.jhydrol.2012.05.036>
- Lucas MC, Kublik N, Rodrigues DB, Meira Neto AA, Almagro A, Melo DDC, Gomes VJC, Pinto LB, Oliveira PTS (2021) Significant baseflow reduction in the São Francisco River Basin. *Water* 13(1):2. <https://doi.org/10.3390/w13010002>
- Maldaner CH (2010) Recarga de aquífero em área urbana: estudo de caso de Urânia (SP) (Aquifer recharge in an urban area: a case study of Urânia [SP]). MSc Thesis, Universidade de São Paulo. <https://doi.org/10.11606/D.44.2010.tde-08012011-201628>
- Manzione RL, Correa AR, Oliveira ADA, do Prado AB, Ferreira JJ, dos Santos LA, Fernandes MR, de Almeida MM, Machado RC G, de Barros RRF (2021) Estimativas da recarga do Sistema Aquífero Bauru-Caiuá entre os anos de 2013 e 2016 (Recharge estimates of the Bauru-Caiuá Aquifer System between 2013 and 2016). *Anais do XX Congresso Bras De Águas Subterrâneas*. <https://doi.org/10.14295/ras.v35i2.30164>
- Menegazzo MC (2016) Sedimentação da Bacia Bauru: cenário tectônico, idade dos depósitos e ambientes paleodeposicionais (Sedimentation of the Bauru Basin: tectonic scenario, age of the deposits and paleodepositional environments). PhD Thesis, Universidade Estadual Paulista. <http://hdl.handle.net/11449/138764>. Accessed 7 Jul 2026
- Moeck C, Grech-Cumbo N, Podgorski J, Bretzler A, Gurdak JJ, Berg M, Schirmer M (2020) A global-scale dataset of direct natural groundwater recharge rates: a review of variables, processes and relationships. *Sci Total Environ* 717:137042. <https://doi.org/10.1016/j.scitotenv.2020.137042>
- Ortiz-Partida JP, Fernandez-Bou AS, Maskey M, Rodríguez-Flores JM, Medellín-Azuara J, Sandoval-Solis S, Ermolieva T, Kanavas Z,

- Sahu RK, Wada Y, Kahil T (2023) Hydro-economic modeling of water resources management challenges: current applications and future directions. *Water Econ Policy* 9(1):2340003. <https://doi.org/10.1142/S2382624X23400039>
- Park E, Parker JC (2008) A simple model for water table fluctuations in response to precipitation. *J Hydrol* 356(3–4):344–349. <https://doi.org/10.1016/j.jhydrol.2008.04.022>
- Paula e Silva F, Chang HK, Caetano-Chang MR (2005) Estratigrafia de subsuperfície do Grupo Bauru (K) no estado de São Paulo (Subsurface stratigraphy of the Bauru Group [K] in the state of São Paulo). *Rev Bras Geoci* 35(1):77–88
- Peres GS, Crioni PLB, Manara CM, Teramoto EH, Kiang CH (2025) Trend analysis in hydrological time series of the Bauru Aquifer System (BAS). *Derbyana* 46:e875. <https://doi.org/10.69469/derb.v46.875>
- Ramires T (2017) Balanço hídrico e recarga do Sistema Aquífero Bauru (SAB) com dados de diferentes resoluções espaciais em Águas de Santa Bárbara-SP (Water balance and recharge of the Bauru Aquifer System [SAB] using data from different spatial resolutions in Águas de Santa Bárbara-SP). MSc Thesis, Universidade Estadual Paulista. <http://hdl.handle.net/11449/150742>. Accessed 7 July 2026
- Renna Camacho C, Getirana A, Rotunno Filho OC, Mourão MAA (2023) Large-scale groundwater monitoring in Brazil assisted with satellite-based artificial intelligence techniques. *Water Resour Res* 59:e2022WR033588. <https://doi.org/10.1029/2022WR033588>
- Rudolph MG, Collenteur RA, Kavousi A, Giese M, Wöhling T, Birk S, Hartmann A, Reimann T (2023) A data-driven approach for modelling Karst spring discharge using transfer function noise models. *Environ Earth Sci* 82(13):339. <https://doi.org/10.1007/s12665-023-11073-1>
- Samani S (2021) Assessment of groundwater sustainability and management plan formulations through the integration of hydrogeological, environmental, social, economic and policy indices. *Groundw Sustain Dev* 15:100681. <https://doi.org/10.1016/j.gsd.2021.100681>
- Scanlon BR, Healy RW, Cook PG (2002) Choosing appropriate techniques for quantifying groundwater recharge. *Hydrogeol J* 10:18–39. <https://doi.org/10.1007/s10040-001-0176-2>
- Scanlon BR, Fakhreddine S, Rateb A, de Graaf I, Famiglietti J, Gleeson T, Grafton RQ, Jobbagy E, Kebede S, Kolusu SR, Konikow LF (2023) Global water resources and the role of groundwater in a resilient water future. *Nat Rev Earth Environ* 4(2):87–101. <https://doi.org/10.1038/s43017-022-00378-6>
- Seidenfaden IK, Mansour M, Bessiere H, Pulido-Velazquez D, Højberg A, Atanaskovic Samolov K, Baena-Ruiz L, Bishop H, Dessì B, Hinsby K, Hunter Williams NH, Larva O, Martarelli L, Mowbray R, Nielsen AJ, Öhman J, Petrovic Pantic T, Stroj A, van der Keur P, Zaadnoordijk WJ (2023) Evaluating recharge estimates based on groundwater head from different lumped models in Europe. *J Hydrol Reg Stud* 47:101399. <https://doi.org/10.1016/j.ejrh.2023.101399>
- Soltanian MR, Moeini F, Dai Z, Sawyer AH, Fleckenstein JH, Doherty J, Curtis Z, Chaudhuri A, Chiogna G, Fahs M, Han WS, Mseli ZH, Lotti F, Moon HS, Zhu L, Al-Masri D, Zhan C, Hoteit H, Matin MA, Zarei A, Carroll KC, Evans SG, Madani K (2024) Sustainability nexus AID: groundwater. *Sustain Nexus Forum* 32(1):21. <https://doi.org/10.1007/s00550-024-00557-7>
- Taghilou AA (2022) A framework resilient social-ecological system (SES) for groundwater governance and public and private rights. *Groundw Sustain Dev* 18:100788. <https://doi.org/10.1016/j.gsd.2022.100788>
- UNESCO (2022) The United Nations world water development report 2022: groundwater: making the invisible visible. UNESCO, Paris
- Zambianqui RP (2018) Monitoramento dos níveis freáticos do Aquífero Bauru (Formação Adamantina) no município de Bastos-SP frente a correlações com o regime de precipitação. MSc Thesis, Universidade Estadual Paulista. <http://hdl.handle.net/11449/194279>. Accessed 7 Jul 2026

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.